

- Description

This module manages a smart factory noise level monitoring system, ensuring that:

1. The system continuously monitors the current noise level (in decibels, dB).
2. If the noise level exceeds a predefined threshold, an alert is triggered indicating that noise levels are too high and may require intervention.
3. If the noise level is below the threshold, the system indicates that the noise is within safe limits.

This system helps maintain a safe and compliant working environment by providing timely alerts when noise levels become excessive.

- Specification

module SmartNoiseMonitoring

var currentNoise, noiseThreshold: real

process updateNoise(newNoise: real) returns (currentNoise: real)

functional scenario form:

currentNoise > noiseThreshold && output "ALERT: Noise level (" + currentNoise + " dB) exceeds threshold (" + noiseThreshold + " dB). Intervention required.", High noise alert triggered.

||

currentNoise <= noiseThreshold && output "Noise level (" + currentNoise + " dB) is within safe limits.", Noise level is acceptable.

end process

end module

- Java Code

```
package code;
public class c104_SmartNoiseMonitor {
    private double noiseThreshold;
    private double currentNoise;
    public c104_SmartNoiseMonitor(double noiseThreshold, double initial
Noise) {
        this.noiseThreshold = noiseThreshold;
        this.currentNoise = initialNoise;
    }
    public void updateNoise(double newNoise) {
        currentNoise = newNoise;
        System.out.printf("Current Noise Level: %.2f dB%n", currentNois
e);
        if (currentNoise > noiseThreshold) {
            System.out.println("ALERT: Noise level too high! Interventi
on required.");
        } else {
            System.out.println("Noise level is within safe limits.");
        }
    }
}
```

```

    }
}
public double getCurrentNoise() {
    return currentNoise;
}
public static void main(String[] args) {
    c104_SmartNoiseMonitor monitor = new c104_SmartNoiseMonitor(85.0,
70.0);
    // testcase-VT:
    //     monitor.updateNoise(75.0);
    //     monitor.updateNoise(90.0);
    //     monitor.updateNoise(80.0);

    // testcase-FT:
    //     monitor.updateNoise(-1.7976931348623157E308);
    //     monitor.updateNoise(-1.0715086069367872E301);
    //     monitor.updateNoise(-2.4850550729069995E291);

    // testcase-Z3:
    //     monitor.updateNoise(75.0);
    //     monitor.updateNoise(90.0);
    //     monitor.updateNoise(80.0);

    // testcase-UVT:
    monitor.updateNoise(75.0);
    monitor.updateNoise(90.0);
    monitor.updateNoise(80.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c104_SmartNoiseMonitor.java	100.0 %	2	0	2
▼ c104_SmartNoiseMonitor	100.0 %	2	0	2
● updateNoise(double)	100.0 %	2	0	2
● c104_SmartNoiseMonitor(doubl		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,850	21,904
▼ src	0.2 %	54	21,850	21,904
▼ code	0.2 %	54	21,850	21,904
▼ c104_SmartNoiseMonitor.java	94.7 %	54	3	57
▼ c104_SmartNoiseMonitor	94.7 %	54	3	57
● updateNoise(double)	100.0 %	29	0	29
● c104_SmartNoiseMonitor(doubl	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.1 %	1	841	842
▼ src	0.1 %	1	841	842
▼ code	0.1 %	1	841	842
▼ c104_SmartNoiseMonitor.java	50.0 %	1	1	2
▼ c104_SmartNoiseMonitor	50.0 %	1	1	2
● updateNoise(double)	50.0 %	1	1	2
● c104_SmartNoiseMonitor(doubl		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 87.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	50	21,854	21,904
▼ src	0.2 %	50	21,854	21,904
▼ code	0.2 %	50	21,854	21,904
▼ c104_SmartNoiseMonitor.java	87.7 %	50	7	57
▼ c104_SmartNoiseMonitor	87.7 %	50	7	57
● updateNoise(double)	86.2 %	25	4	29
● c104_SmartNoiseMonitor(doubl	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c104_SmartNoiseMonitor.java	100.0 %	2	0	2
▼ c104_SmartNoiseMonitor	100.0 %	2	0	2
● updateNoise(double)	100.0 %	2	0	2
● c104_SmartNoiseMonitor(doubl		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,850	21,904
▼ src	0.2 %	54	21,850	21,904
▼ code	0.2 %	54	21,850	21,904
▼ c104_SmartNoiseMonitor.java	94.7 %	54	3	57
▼ c104_SmartNoiseMonitor	94.7 %	54	3	57
● updateNoise(double)	100.0 %	29	0	29
● c104_SmartNoiseMonitor(doubl	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c104_SmartNoiseMonitor.java	100.0 %	2	0	2
▼ c104_SmartNoiseMonitor	100.0 %	2	0	2
● updateNoise(double)	100.0 %	2	0	2
● c104_SmartNoiseMonitor(doubl		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 94.7%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool		0.2 %	54	21,850	21,904
▼ src		0.2 %	54	21,850	21,904
▼ code		0.2 %	54	21,850	21,904
▼ c104_SmartNoiseMonitor.java		94.7 %	54	3	57
▼ c104_SmartNoiseMonitor		94.7 %	54	3	57
● updateNoise(double)		100.0 %	29	0	29
● c104_SmartNoiseMonitor(doubl		100.0 %	9	0	9
● main(String[])		100.0 %	16	0	16